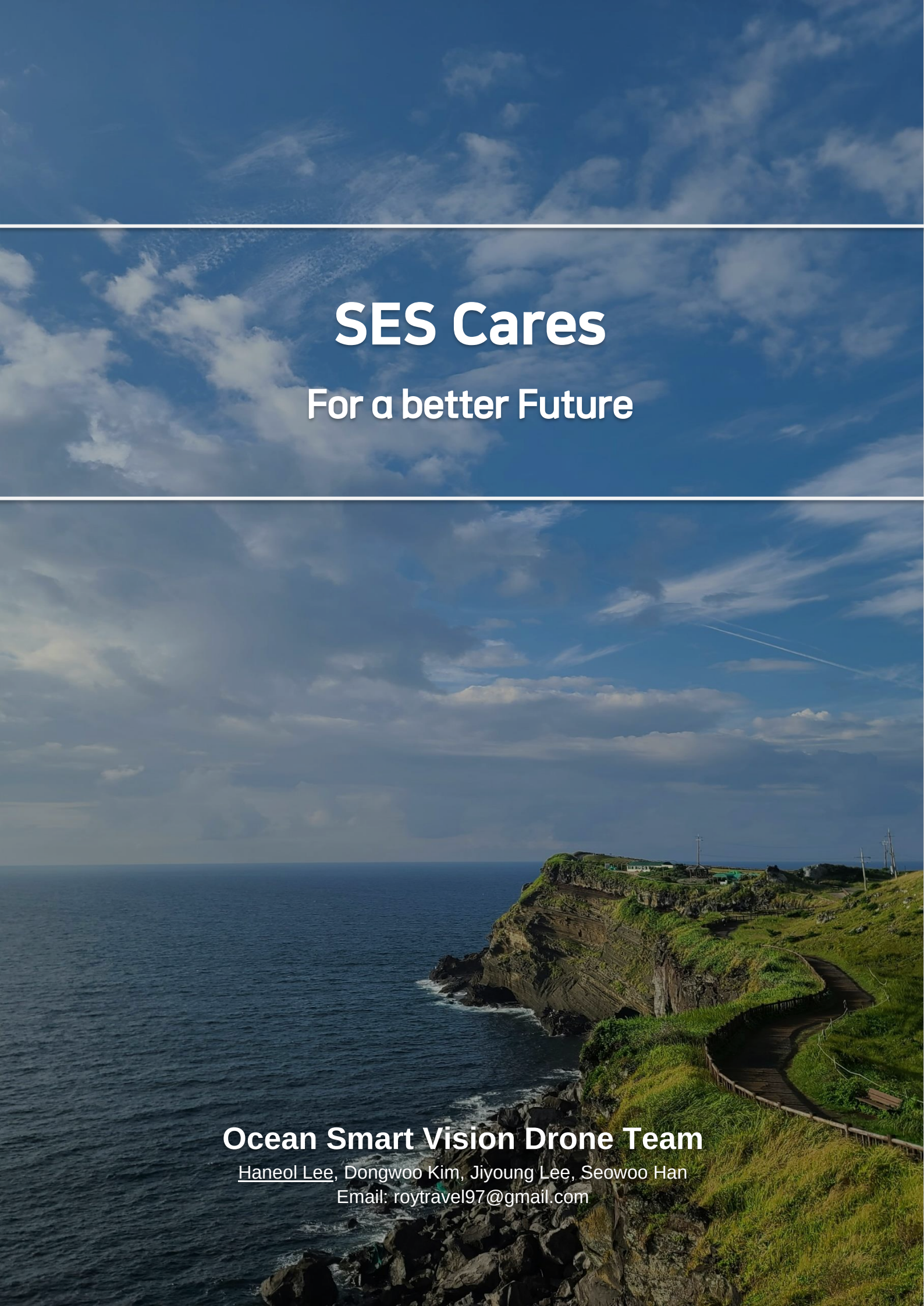




SES Cares

For a better Future

Ocean Smart Vision Drone Team

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1. Abstract

Driven by the surge in marine waste and the need to protect endangered animals, this research proposes to utilize innovative drone technology to effectively respond to environmental issues and protect endangered species such as the southern bottlenose dolphin. Habitat surveillance of endangered animals using drones can help us quickly identify animal health conditions and threats from human activities so that we can take necessary protective measures. The Endangered Habitat Surveillance project is being implemented directly in 2023, and we believe that we can build on this pipeline to conduct intelligent marine waste monitoring, which is expected to contribute to the preservation of sustainable marine ecosystems by providing a real-time view of the marine environment and helping us prepare efficient cleanup strategies. These two projects overcome the limitations of traditional monitoring methods and present a groundbreaking approach to environmental issues and supporting vulnerable populations through modern technological innovation. We want to solve this problem by focusing on a region in South Korea called Jeju Island, which is closest to where we live.

2. Problem Statement

2.1 Jeju South Bottlenose Dolphin

The endangered Jeju southern bottlenose dolphin is currently facing restrictions on its habitat and ecological activities due to tourist vessels. To preserve the diversity of marine life, ¹Article 22, paragraphs 1 and 2 of South Korea's Marine Ecosystem Act prohibits tourist vessels from excessively approaching marine protected species or operating at high speeds.

CONSERVATION AND MANAGEMENT OF MARINE ECOSYSTEMS ACT	
Article 22 (Restrictions on Advertisements and Observation Activities relating to Marine Organisms under Protection)	
(1)	No person shall place an advertisement which is likely to accelerate the extinction or decrease, or provoke the abuse, of marine organisms under protection: <i>Provided</i> , That this shall not apply where he or she receives authorization or permission, etc. under other statutes. (Amended on Dec. 31, 2018; Oct. 18, 2022)
(2)	Any person who intends to perform observation or tourism activities in relation to marine organisms under protection shall not conduct any of the following acts that are likely to interrupt or disturb the migration or feeding activities of marine organisms under protection: <i>Provided</i> , That marine organisms under protection in zoos and aquariums prescribed in the Act on the Management of Zoos and Aquariums shall be governed by such Act: (Newly Inserted on Oct. 18, 2022)
1.	Approaching marine organisms under protection too closely;
2.	Operating ships, etc. at a higher than prescribed speed;
3.	Feeding marine organisms under protection at will;
4.	Any other activities prescribed by Ordinance of the Ministry of Oceans and Fisheries, which are likely to interfere with the migration or feeding activities of marine organisms under protection.
(3)	Matters necessary for the detailed standards, methods, etc. for the observation of marine organisms under protection or tourism activities involving them under paragraph (2) shall be prescribed by Ordinance of the Ministry of Oceans and Fisheries. (Newly Inserted on Oct. 18, 2022)
[Title Amended on Oct. 18, 2022]	

Figure 1. Article 22 of Conservation And Management Of Marine Ecosystems Act

In particular, according to the 'Detailed Standards and Methods for Marine Protected Species Observation or Tourism Activities', the distance between the vessel and the dolphin should be at least 50 meters, and the speed of the vessel should be observed according to the distance from the dolphin, as shown below.

¹ https://elaw.klri.re.kr/kor_service/lawView.do?hseq=61419&lang=ENG

Distance from dolphin	1,500m ~ 750m	750m ~ 300m	300m ~ 50m
Vessel speed	Less than 10 knots (18.52km/h)	Less than 5 knots (9.26km/h)	Stop vessel screw

Table 1. Vessel speed regulations based on distance from dolphins

The problem, however, is that despite these laws being in place, tourist vessels continue to operate without complying with them. To find out, we interviewed experts from the Marine Animal Research & Conservation (MARC) organization, Sujin Jang and Miyeon Kim.

MARC said that in order to prepare countermeasures against the illegal activities of these tourist ships, they filmed the activities with a drone and submitted the video to the Jeju Provincial Government. However, they received a response from the Jeju Provincial Government that it is difficult to punish the ships because the distance and speed of the ships cannot be quantitatively measured based on the video alone. In other words, although the Marine Ecosystem Act restricts tourism activities for the sake of marine protected species, the law is not effective in practice.

Solving these blind spots in the law is a problem that must be solved to maintain the balance and stability of the marine ecosystem, preserve its diversity, and leave a better world for future generations. As an idea to solve this problem, we propose an AI-based drone image analysis system. The AI-based video analysis system will quantitatively estimate the distance between the ship and dolphins and the speed of the ship to determine whether the tourist vessel is in violation of the law and provide the Jeju Island government with a basis for disposing of fines.

To discuss the feasibility of this idea in practice and whether there are any further considerations, our team, along with MARC, conducted a meeting with Jieun Lee and Seungcheol Lee of the Maritime Industry Division of the Jeju Provincial Office of Fisheries and Oceans. As a result of the interview, we learned that the Jeju Provincial Government currently uses advisory personnel such as the pilots' association and video/drone experts to visually determine the distance, speed, and screw stops in the video to determine violations. However, there are ambiguous situations that prevent them from considering punishment, and if an AI-based drone video analysis system is developed, they would like to adopt it as a basis for punishment.

Based on this response, we decided that the development of an AI-based drone video analysis system could play a key role in solving the problem of blind spots in the Marine Ecosystem Act, and we implemented the system ourselves in 2023, created an MVP, and submitted it to Jeju Island with MARC by analyzing the video of illegal sites with AI.

2.2 Jeju Marine Waste

The amount of marine waste is currently skyrocketing around the world. A study by the Algalita Marine Research Foundation found that plastic waste in the Central Pacific Gyre increased fivefold from 1997 to 2007. Initial results from 1997 showed that plastic debris on the ocean surface outnumbered plankton, a plant-based animal, by a ratio of 6:1.4. Along with endangered marine species, plastic waste on the ocean surface is a major challenge that must be addressed for a sustainable marine environment.

In particular, the Sanbuk coast of Jeju City and the Sannam coast of Seogwipo City are intensively affected by marine debris due to the effects of northwest monsoons in January and typhoons in summer, respectively. The Korea Marine Environment Corporation has been conducting national coastal debris monitoring every two months since 2018 with the support of the Ministry of Oceans and Fisheries, and according to the monitoring report, 60% of the total weight of marine debris generated in each region over the course of a year is concentrated in January for the Gimnyeong coast of Jeju City and a whopping 80% in July for the Sagye coast of Seogwipo City. The Ministry of Oceans and Fisheries is trying to collect marine debris at all times by deploying marine environmental guards, but there is a problem of insufficient manpower compared to the amount of debris.

In addition, if you look at the data by type of marine debris monitoring, the classification of debris varies from glass, plastic, and metal, and even within the same plastic, there are countless types, from fiber to film. If the amount of garbage by type is provided in advance, resources and manpower can be deployed and managed more efficiently. Therefore, we would like to propose an AI-based drone image analysis system. If the AI-based drone image analysis system is used to detect marine debris accumulated along the coastline, and specific information such as the type, amount, and size of marine debris is continuously obtained, it will be possible to more efficiently deal with marine debris that is concentrated by season in each region.

3. Goals

1) Secure policy recognition of AI-based drone image analysis system and propose law amendment

- **Purpose:** To contribute to the protection of marine protected species and preservation of diversity to ensure a sustainable marine ecosystem for future generations.

- **Goal:** The project will develop an AI-based drone video analysis system that can quantitatively analyze the distance between ships and dolphins, ship speed, etc. to determine whether tourist vessels are in violation and use it as a basis for administrative penalties.

- **Method:** Consult with the National Assembly and the Ministry of Oceans and Fisheries to secure policy credibility, and propose amendments to the current law so that the system can be legally recognized.

2) Develop accurate and efficient marine waste collection methods

- **Purpose:** To protect Sagyeri's marine environment by enhancing waste collection efficiency, promoting sustainable conservation.

- **Goal:** Utilize drone technology for advanced marine waste monitoring, enabling strategic manpower and resource allocation based on real-time waste data analysis.

- **Method:** By mapping marine waste movement and analyzing drone-captured footage, we'll develop targeted collection strategies, optimizing cleanup efforts and equipment deployment for varied waste types and sizes.

4. Project

The project aims to utilize AI-powered drones for marine endangered species conservation and marine waste monitoring. First, for endangered animals such as the Jeju southern bottlenose dolphin, drones regularly patrol their natural habitat to monitor their living patterns, breeding activities, and potential threats from human activities. The data they collect is then analyzed by AI to monitor the animals' health and habitat changes in real time, providing information that can be used to take appropriate protective measures if necessary. In the case of this project, our team personally conducted video analysis based on footage captured by a DJI drone on Jeju Island. You can read more about this in 5.1

During our first project, we saw a lot of marine waste on Jeju Island, and we thought it would be a good idea to film and analyze it at the same time. To address the problem of marine waste, especially plastic pollution, drones identify and locate waste on the ocean surface and along the coastline. AI technology is used to automatically identify and categorize waste from the images and data collected. This helps to identify the main sources of pollution and where the waste is traveling, which provides an important basis for the formulation and implementation of marine waste management strategies.

This approach is much more efficient than traditional monitoring methods and provides the ability to survey large areas quickly. In particular, the SES AI will allow for longer periods of time to better observe and protect Jeju's southern right whales, as well as to capture marine debris in a wider area of the coastline. By leveraging the power of artificial intelligence (AI), the analysis and interpretation of the data collected by the drone will be automated, enabling faster and more accurate decision-making. This project opens up new avenues for the protection and sustainable management of marine ecosystems, and demonstrates a modern approach to addressing environmental challenges through technological innovation.

5. Methods

5.1 Using AI, drones to protect marine endangered species

In order to develop a drone video analysis system, we first need actual drone footage of ships and dolphins, which we were able to obtain from the Marine Animal Ecology and Conservation Institute, which provided us with drone footage from various locations in Jeju over a period of time. Figure 2 is an example image of the footage we received, and if you check the image, you can see the ship and a group of dolphins.

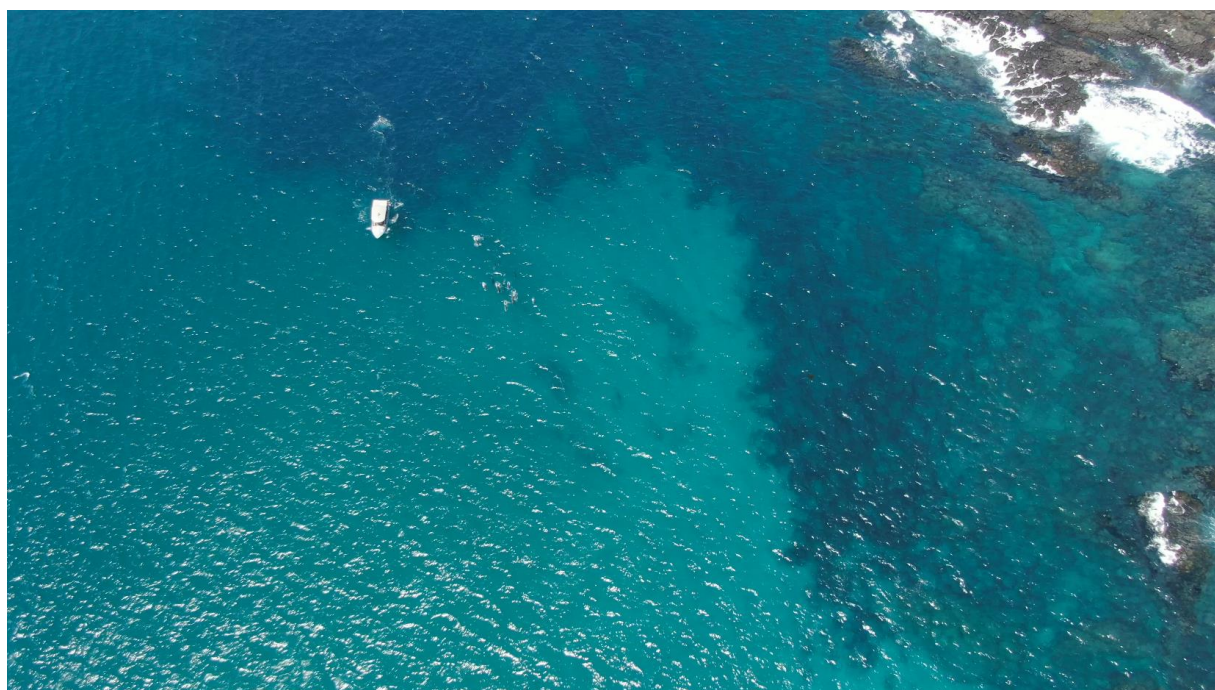


Figure 2. Example of drone video provided by the MARC

When we received the footage, there were two main issues we faced: the first was that there were times when it was difficult to distinguish between dolphins and reflections due to light reflections on the ocean surface. For example, in the footage we received, it was difficult to distinguish between reflections and dolphins as shown below.



Figure 3. Light reflection on sea surface

This problem of light reflections caused the AI model to incorrectly detect reflected light as a dolphin, even when training in real-world conditions. To solve this problem, we realized the need to develop a light reflection removal module. The Light Reflection Removal module reduces light reflections by averaging the brightness between frames over consecutive frames.

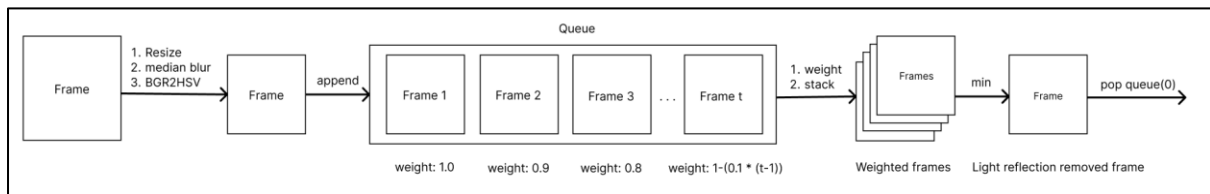


Figure 4. Frame fusion method to remove light reflection

When we have several consecutive frames, we add them to a queue of some length. The frames in the first queue to remove light reflections are weighted with a weight of 1, decreasing by 0.1 each time the next frame is added. The frames are then overlapped and fused by applying the smallest value to each pixel position in the frames. The first frame after the fusion is then dequeued and a new frame is added, and so on.

This removes bright areas, such as light reflections on the ocean surface, while the darker dolphins are less affected, making them a little more visually distinct. The result of applying these light reflection removal modules to the original footage is shown in Figure 5, and we can see that the AI model's detection performance increases when the light reflection removal modules are applied.

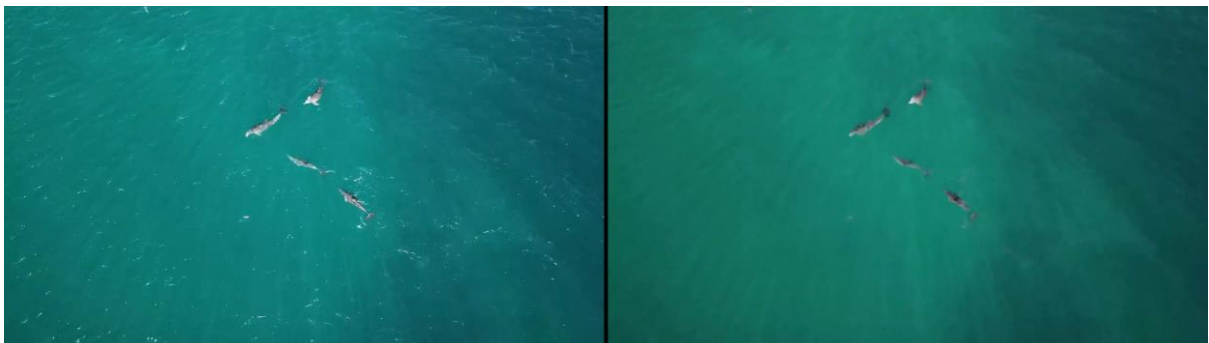


Figure 5. Before/after applying the light reflection removal method

The second challenge we faced, besides the light reflection issue, was the question of whether the dolphins would be able to discern the blurred shape of the dolphins at the surface of the water. For example, we researched previous study[1] and found that dolphins swimming in the ocean appear blurry, as shown below, which reduces the detection reliability of AI models and makes it difficult to make accurate judgments.

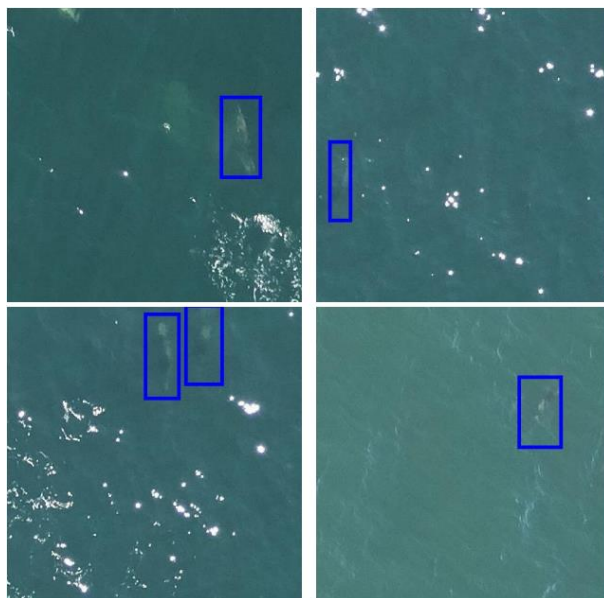


Figure 6. Blurry dolphin shape in the sea surface

The reason for the poor detection reliability of these AI models is the ambiguity that occurs when labeling blurry dolphins during the dataset construction phase. To compensate for this detection reliability issue, we applied a method that only labels dolphin objects when they have a clear shape, except in cases where the dolphin object is somewhat ambiguous at the time of labeling, which would lead to false positives in later training.

The AI model we used for training is the ByteTrack model, which is used for object tracking. It is a model that compensates for the limitations of the DeepSort model, which was the first model designed for object tracking, and greatly improves performance compared to the DeepSort model by not stopping tracking objects even if they are not detected in some frames, but instead waiting for a certain period of time (frames) and tracking them again if they appear in later frames. Here is a visualization of the object tracking results after training with the ByteTrack model.



Figure 7. ByteTrack model inference result

After training, we found that the model was able to detect small objects such as dolphins with high performance. In addition to the ByteTrack model, we also used YOLOv8, an object detection model, and EfficientAD, an anomaly detection model, to reduce false positives by removing the bounding boxes for overlapping objects from the object detection results and anomaly detection results.

To model the actual ship–dolphin distance and the distance between pixels in the image, we used Ground Sample Distance (GSD), which is the distance between the centers of consecutive pixels in an ortho image, as shown in the figure below.

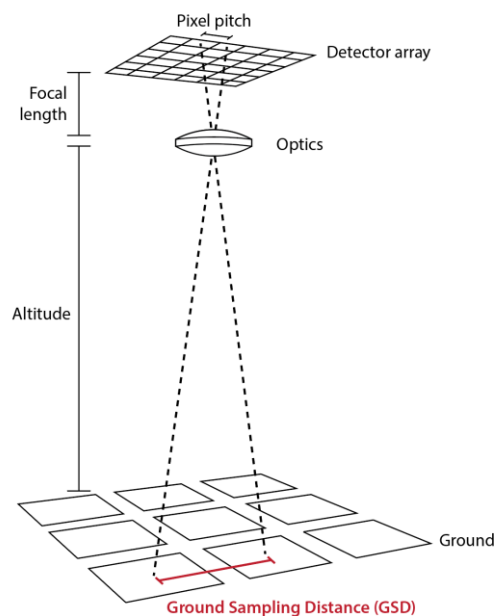


Figure 8. Ground Sample Distance

The key variable that determines the ground sample distance is the altitude of the drone, which is a key factor in surveying with drones because it allows for a more precise calculation of the actual distance. Here, the ground sample distance is calculated using the following formula:

$$GSD_h = \frac{A \times S_h}{F \times I_h}, GSD_w = \frac{A \times S_w}{F \times I_w}$$

A = Altitude of flight

S_h = Height of sensor

S_w = Width of sensor

F = Focal length

I_h = Height of Image

I_w = Width of Image

To calculate the ground sample distance, we need not only altitude information, which is a key variable, but also focal length, image sensor size, image size, etc. to perform georeferencing to find the actual distance, and to obtain the image converted to Bird Eye View, we need information such as camera position, orientation, lens distortion, etc. These information can be obtained by extracting the logs collected on the drone aircraft. The drones we used to collect the footage were the DJI Mavic 2 Zoom and DJI Mavic Pro, and DJI's drones generated two types of logs.

The first was a log in SRT format that was generated from the time the video was taken, and the second was a log in CSV format that was generated during the entire time the drone was in motion. The CSV format log had a "CAMERA.isVideo" column to identify when the video was taken, and we used this as a reference point to combine the two logs by matching the time bands of the two logs and extracting the altitude and focal length information of the drone. We also extracted the roll, pitch, and yaw information expressed in Eulerian coordinates of the drone's attitude and the attitude of the gimbal camera, which we needed to apply Bird Eye View to the drone footage later. An example of a combined drone log with all of this information is shown in the following figure.

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R
FrameCnt	focal_length	datetime	OSD.latitude	OSD.longitude	OSD.xSpeed [MPH]	OSD.ySpeed [MPH]	OSD.zSpeed [MPH]	OSD.directionOfTravel	OSD.pitch	OSD.roll	OSD.yaw	GIMBAL	GIMBAL	GIMBAL	HOME	HOME	adjust
108	107	290	2023-11-26 17:06	33.26366342	126.1792569	0	0	0	43.7	-1.86666667	0.266667	-10.4	-89.9	0	-11.5	126.1825	33.26075 79.4
109	108	290	2023-11-26 17:06	33.2636634	126.1792569	0	0	0	43.7	-1.9	0.3	-10.4	-89.9	0	-11.5	126.1825	33.26075 79.4
110	109	290	2023-11-26 17:06	33.26366343	126.1792569	0	0	0	43.7	-2.76666667	0.3	-10.4333	-89.9	0	-11.5	126.1825	33.26075 79.4
111	110	290	2023-11-26 17:06	33.26366345	126.1792569	0	0	0	43.7	-3.63333333	0.3	-10.4667	-89.9	0	-11.5	126.1825	33.26075 79.4
112	111	290	2023-11-26 17:06	33.26366347	126.1792569	0	0	0	43.7	-4.5	0.3	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
113	112	290	2023-11-26 17:06	33.26366347	126.1792569	0.07456467	0	0	43.7	-8.3	0.266667	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
114	113	290	2023-11-26 17:06	33.26366347	126.1792569	0.149129333	0	0	43.7	-12.1	0.233333	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
115	114	290	2023-11-26 17:06	33.26366347	126.1792569	0.223694	0	0	43.7	-15.9	0.2	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
116	115	290	2023-11-26 17:06	33.26366352	126.1792569	0.372823333	0	0	144.766667	-18.066667	0.2	-10.4667	-89.9	0	-11.5	126.1825	33.26075 79.4
117	116	290	2023-11-26 17:06	33.26366356	126.1792569	0.521952667	0	0	245.8333333	-20.23333333	0.2	-10.4333	-89.9	0	-11.5	126.1825	33.26075 79.4
118	117	290	2023-11-26 17:06	33.26366361	126.179257	0.671082	0	0	346.9	-22.4	0.2	-10.4	-89.9	0	-11.5	126.1825	33.26075 79.4
119	118	290	2023-11-26 17:06	33.26366376	126.1792569	0.969340667	-0.07456467	0	232.366667	-22.966667	0.166667	-10.4333	-89.9	0	-11.5	126.1825	33.26075 79.4
120	119	290	2023-11-26 17:06	33.26366392	126.1792569	1.267599333	-0.149129333	0	117.8333333	-23.53333333	0.133333	-10.4667	-89.9	0	-11.5	126.1825	33.26075 79.4
121	120	290	2023-11-26 17:06	33.26366407	126.1792569	1.565858	-0.223694	0	3.3	-24.1	0.1	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
122	121	290	2023-11-26 17:06	33.26366432	126.1792569	1.864116667	-0.298258667	0	2.533333333	-24.166667	0.1	-10.4667	-89.9	0	-11.5	126.1825	33.26075 79.4
123	122	290	2023-11-26 17:06	33.26366457	126.1792568	2.162375333	-0.372823333	0	1.766666667	-24.23333333	0.1	-10.4333	-89.9	0	-11.5	126.1825	33.26075 79.4
124	123	290	2023-11-26 17:06	33.26366482	126.1792568	2.460634	-0.447388	0	1	-24.3	0.1	-10.4	-89.9	0	-11.5	126.1825	33.26075 79.4
125	124	290	2023-11-26 17:06	33.26366521	126.1792567	2.758892667	-0.521952667	0	1.766666667	-24.2	0.1	-10.4	-89.9333	0	-11.5	126.1825	33.26075 79.4
126	125	290	2023-11-26 17:06	33.26366556	126.1792567	3.057151333	-0.596517333	0	2.533333333	-24.1	0.1	-10.4	-89.9667	0	-11.5	126.1825	33.26075 79.4
127	126	290	2023-11-26 17:06	33.26366598	126.1792566	3.35541	-0.671082	0	3.3	-24	0.1	-10.4	-90	0	-11.5	126.1825	33.26075 79.4
128	127	290	2023-11-26 17:06	33.26366634	126.1792565	3.579104	-0.671082	0	3.366666667	-23.93333333	0.133333	-10.4333	-90	0	-11.5	126.1825	33.26075 79.4
129	128	290	2023-11-26 17:06	33.26366667	126.1792564	3.802798	-0.671082	0	3.433333333	-23.86666667	0.166667	-10.4667	-90	0	-11.5	126.1825	33.26075 79.4
130	129	290	2023-11-26 17:06	33.26366706	126.1792563	4.026492	-0.671082	0	3.5	-23.8	0.2	-10.5	-90	0	-11.5	126.1825	33.26075 79.4
131	130	290	2023-11-26 17:06	33.26366765	126.1792562	4.324750667	-0.745646667	0	3.466666667	-23.76666667	0.233333	-10.5	-89.9667	0	-11.5	126.1825	33.26075 79.4
132	131	290	2023-11-26 17:06	33.26366825	126.1792561	4.623009333	-0.820211333	0	3.433333333	-23.73333333	0.266667	-10.5	-89.9333	0	-11.5	126.1825	33.26075 79.4
133	132	290	2023-11-26 17:06	33.26366884	126.1792559	4.921268	-0.894776	0	3.4	-23.7	0.3	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
134	133	290	2023-11-26 17:06	33.26366951	126.1792558	5.144962	-0.969340667	0	3.766666667	-23.73333333	0.266667	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
135	134	290	2023-11-26 17:06	33.26367018	126.1792556	5.368656	-1.049905333	0	4.133333333	-23.76666667	0.233333	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
136	135	290	2023-11-26 17:06	33.26367086	126.1792554	5.59235	-1.11947	0	4.5	-23.8	0.2	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4
137	136	290	2023-11-26 17:06	33.26367185	126.1792552	5.816044	-1.11947	0	4.166666667	-23.76666667	0.233333	-10.5	-89.9	0	-11.5	126.1825	33.26075 79.4

Figure 9. Example of combined drone flight log

One key consideration is that the drone altitude in the drone logs represents the distance from the ground where the flight started, so we needed to convert this to the drone altitude at sea. This means that to determine the drone altitude at sea, we need to know the difference in altitude between the ground and the sea, which can be calculated by the figure below.

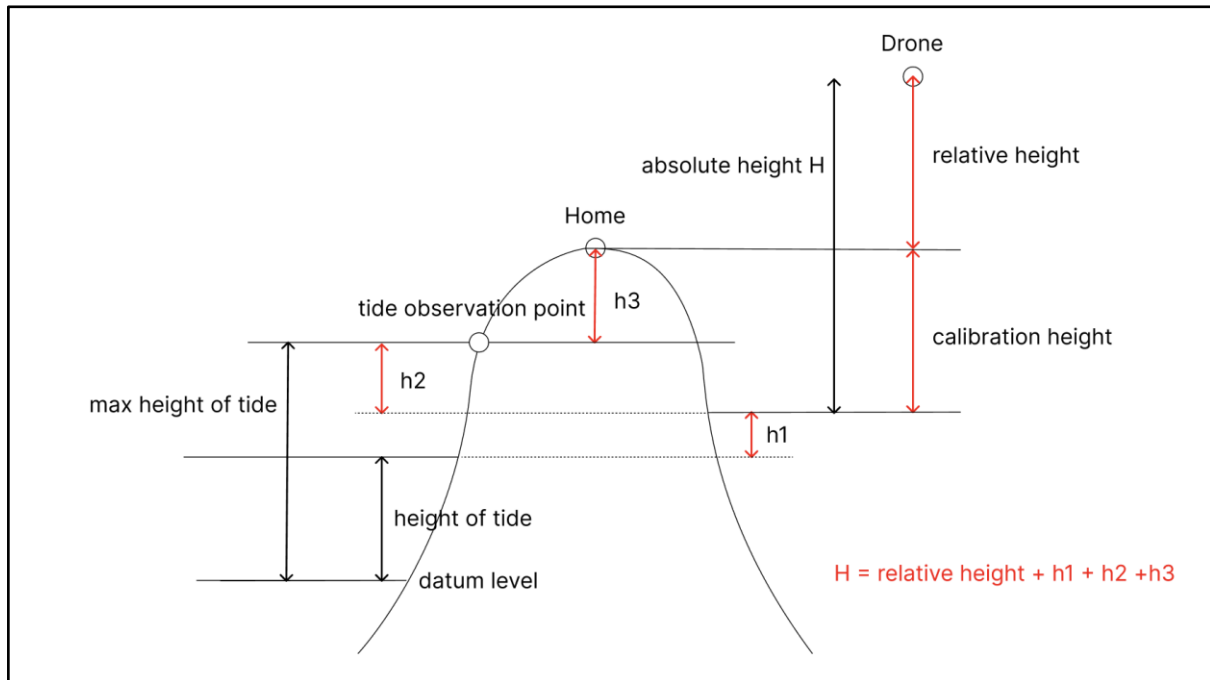


Figure 10. Drone altitude estimation at sea

Based on the figure above, the altitude of the drone at sea can be calculated by the following formula

Drone altitude at sea = (Relative altitude) + $h1$ + $h2$ + $h3$. The relative altitude information in the GPS information was extracted from the drone flight log, and the tide observation value was obtained from the tide station actual and predicted tide information provided by the National Ocean Survey Institute of the Ministry of Oceans and Fisheries in the form of Open API. In addition, the geoid model of the Korea National Geospatial-Intelligence Agency was utilized to obtain the geoid altitude information. The ground sample distance can be derived by calculating the drone altitude information at sea through this information.

Measuring the actual distance through the ground sample distance is based on the orthoimage converted to Bird Eye View. Therefore, a quadrature conversion process is required to convert the given drone image to Bird Eye View. The quadrature conversion mainly applies the perspective or orthographic method, and we utilized Orthophoto Maps, an open source that can apply orthographic to drone images, to perform the quadrature conversion to calculate the ship-dolphin distance. Here is an example image of the resulting visualization of the Bird Eye View by performing a quadrature transformation on the drone footage.

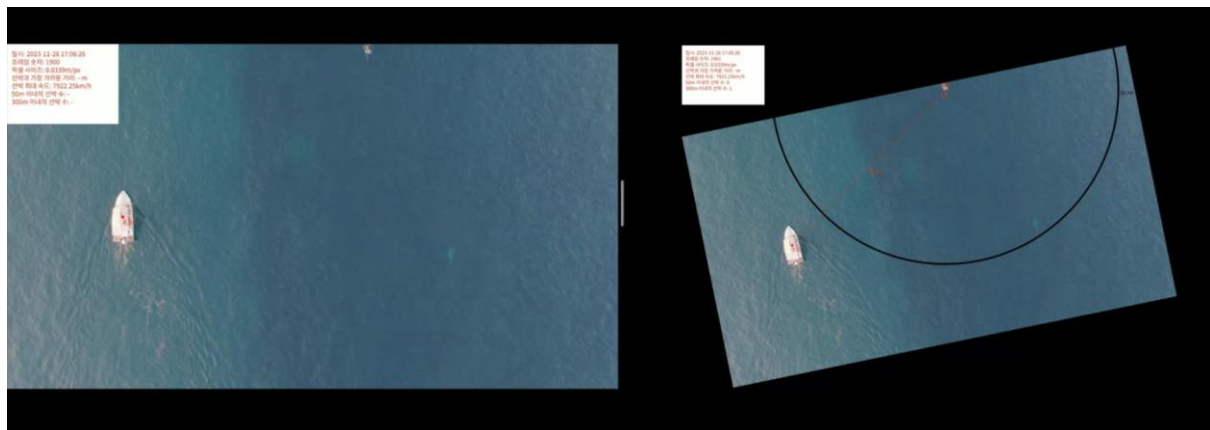


Figure 11. Before/after applying Bird Eye View

In practice, we measured the ship-dolphin distance and applied a visualization to intuitively determine whether it was illegal or not in order to submit it to the Jeju Island government as a video of illegal vessel detection.



Figure 12. Visualization of ByteTrack model inference results (legal)



Figure 13. Visualization of ByteTrack model inference results (illegal)

For the dolphins, we merged the bounding boxes for the visualization, and when measuring the distance between the ship and the dolphins, we measured the distance between the center of the pod and the ship. The reason for this is that sometimes multiple tourist ships approach the dolphins, and depending on the position of the individual ships, the dolphin that is the center of the distance calculation will change, so we want to make sure that the measurements are consistent for all cases depending on the position of the ships.

5.2 Using AI, drones to monitor marine waste

Marine waste monitoring technology can be implemented in the form of a process that involves flying a drone over a set route at specific times, collecting footage, and using the collected footage to detect marine waste.

Acquiring video data

Drones fly along predefined routes and continuously acquire images of the marine environment. This enables real-time monitoring of a wide range of marine areas and plays a key role in tracking the distribution and evolution of marine waste. SES's battery technology is expected to synergize with the following flight scenarios, as well as the acquisition of images over long periods of time while flying along a predefined path.

[Flight Scenario]

- Flight Site & Route : Sagyeri Coast ~ Yongmuri Coast, Andeok-myeon, Seogwipo-si, Jeju-do => 2,134.3m

- Drone Specifications : SES AI
- Flight Altitude : Flight above the minimum altitude to calculate GSD

Refining and analyzing video data

We found lots of papers about this work. We can get insight from M Wolf et al [2]. We make an algorithm for refining and analyzing drone video data. To refine the acquired image data, we perform pre-processing to improve resolution and remove noise. Then, to detect marine waste, we use Yolo v8, an object detection algorithm, to identify marine waste (Classification) and precisely locate it (Localization).

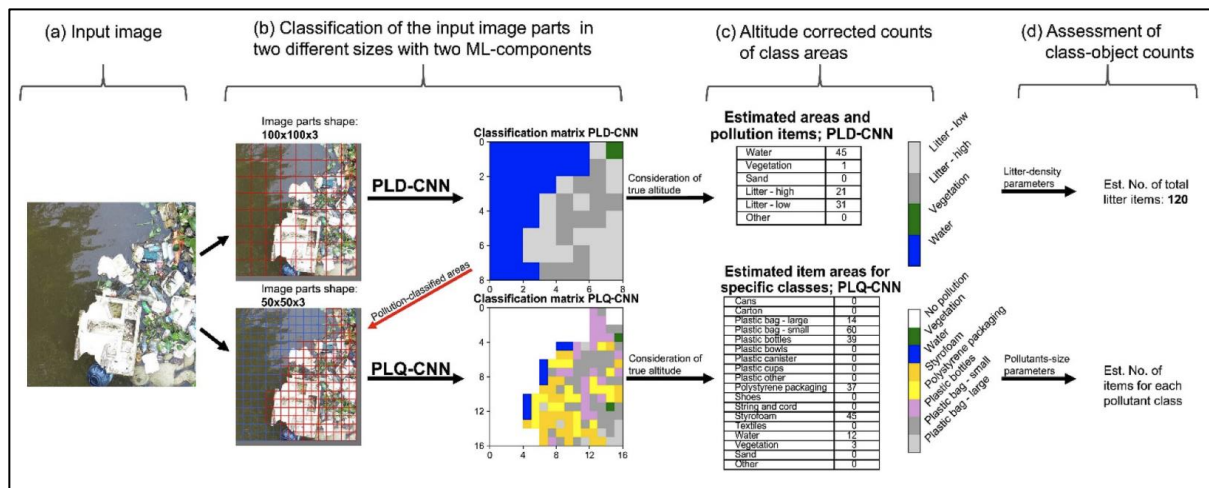


Figure 14. Example for classification waste using drone image (M Wolf et al.)

Collection Strategy and estimation the size and location of marine waste

The logs and metadata acquired during drone filming can be converted into video images in Bird Eye View to obtain the ground sample distance (GSD) to estimate the approximate size of the waste, which can be used to suggest effective methods for waste collection.

In addition, the coordinates of the waste can be estimated from the location coordinates of the drone taking the video, providing data for more effective collection.

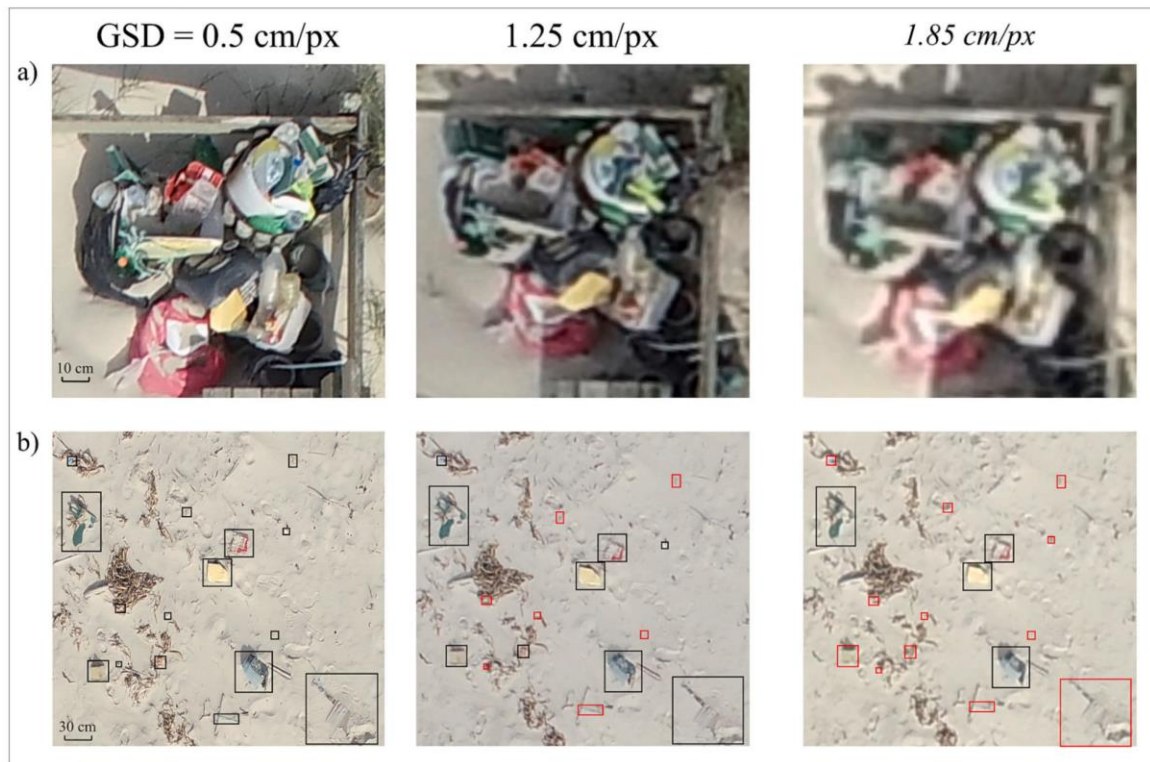


Fig. 3. Example of drone images with different Ground Sampling Distance (GSD). a) aerial picture of a recollection point close to the beach, with bins and garbage bags. Different types of plastic bottles are distinguishable on the image with GSD = 0.5 cm/px (left). On the image with GSD = 1.25 cm/px (central), items can still be counted but not recognized, whereas it is difficult to count litter on the image with GSD = 1.85 cm/px (right); b) operational assessments of litter items marked on images. On the image with GSD = 0.5 cm/px (left), several items were detected, and their material and/or category can be recognized (black boxes). On the image with GSD = 1.25 cm/px (central), the smallest items could not be detected (red boxes), while only the four biggest items could be spotted on the image with GSD = 1.85 cm/px (right). Overall, in b), black boxes bound the detected items, whereas red boxes bound the missed items. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

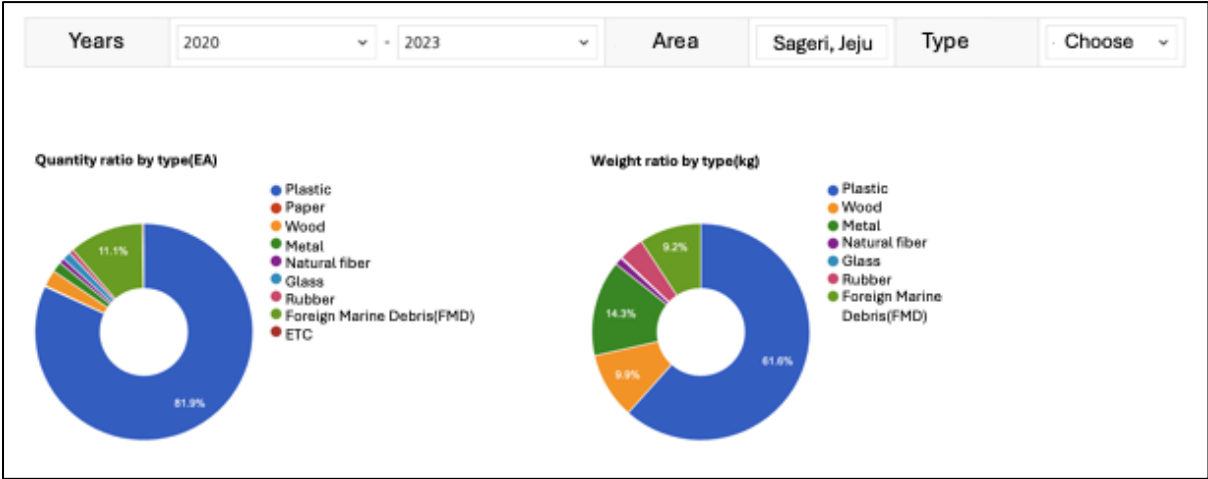
Figure 15. Example of drone images with different GSD (U. Andriolo et al.)

6. Impact Assessment

The marine plastic waste problem continues to have a serious impact on the environment and ecosystems, resulting in a wide range of plastic waste of varying sizes, from microplastics (less than 5 mm in diameter) to macroplastics (greater than 25 mm in diameter). These plastic wastes gradually convert to microplastics through photodegradation, corrosion, and weathering, posing a serious risk to marine life, as well as potentially affecting humans through the food chain. Given the severity of these issues, it is imperative to develop an efficient management and collection strategy for marine waste.

Recent advances in drone technology are providing a new tipping point in marine waste management. In particular, marine waste detection using unmanned aerial vehicles (UAVs) has the potential to revolutionize waste monitoring and identification over large coastlines and marine areas. By enabling the accurate location and size classification of marine debris, these technologies can play an important role in the rapid collection of large debris and the prevention of microplastics accumulation, particularly in the case of oversized debris.

According to the Marine Waste Statistics Portal (<https://www.meis.go.kr/mli/monitoringInfo/stat.do>), plastic waste accounted for the largest proportion of marine waste collected in Sageri, Jeju Island from 2020 to 2023, which has a negative impact not only on the marine ecosystem but also on the marine economy. The wide distribution and diversity of plastic waste presents a major challenge for collection and management, and the introduction of an AI monitoring system utilizing drones is expected to help accelerate problem solving.



The detection of marine debris by drones can provide data for the establishment of marine protected areas, accurate identification of the type and amount of marine debris, and marine ecosystem restoration projects. This can significantly increase the effectiveness of marine protection policies, as well as make a significant contribution to improving the health of marine ecosystems and protecting marine biodiversity. The technology is also expected to facilitate the development of sustainable management and collection strategies for marine waste.

Overall, drone-assisted marine waste detection technology has the potential to become an innovative tool in marine protection and management, which can be considered an important development for the conservation of marine ecosystems and sustainable management of the marine environment. The development and application of such technologies can raise awareness of marine waste issues and become part of international efforts to protect the marine environment.

Based on the statistics of the annual collection of marine waste in Korea, it can be inferred that the proportion of coastal waste is considerable, and that the AI marine waste monitoring system utilizing drones will significantly help preserve marine ecosystems.

In addition, through the Bird Eye View conversion technology that we already possess, we are able to shoot from a wide field of view, breaking away from the vertical shooting technique that has been the subject of control in most studies, and it is expected that we will be able to expand the technology to not only detect marine debris but also to monitor the state of the ecosystem in the situation, thus making an impact as a technology that is not limited to the marine debris problem.

AI-based drone image analysis systems can help protect marine life habitats and maintain ecological balance by monitoring illegal activities on tourist vessels. In addition, in response to marine litter, drones can effectively detect and locate waste moving along the ocean surface and coastline. This is expected to not only maintain the stability and diversity of marine ecosystems, but also enhance the response to waste polluting the marine environment.

On the social side, AI-powered drone systems are expected to raise social awareness of marine conservation issues and emphasize the importance of environmental protection. By addressing legal loopholes and encouraging tourism vessels to comply with norms, they are also expected to emphasize social responsibility for marine conservation, and responsible operations by tourism companies are expected to lead to improved tourism. These collective impacts are expected to contribute to a sustainable environment and improved social life.

In addition, we have submitted a paper to the 2024 IEEE Conference on Artificial Intelligence, which is currently under peer review, on the results of our research on marine endangered species protection using AI-based drones. If the research results lead to the publication of a paper, it will be a great contribution in that it will secure credibility for future revisions of current laws, raise social awareness of these environmental and ecological issues, and open the way for further research.

7. Conclusion

The AI-based drone video analysis system can quantitatively estimate the ship-dolphin distance and ship speed, which is recognized as an effective tool for the Jeju Island government to quickly determine whether a tourist vessel is in violation of the law and use it as a basis for deciding on fines. We are already in the process of utilizing the system to detect illegal vessels and forwarding the results to the Jeju Provincial Office for administrative action. If adopted, this system will be the first case of its kind to address the problem of legal blind spots and marine ecosystem conservation, and the accumulation of such success stories will be used as a basis for the National Assembly to propose amendments to the current law.

The system could also be applied to marine debris monitoring, where drones could be utilized to efficiently detect and track waste moving along the ocean surface and coastline. This will help to continuously monitor the marine environment and identify sources of pollution. These efforts are expected to contribute to maintaining the stability and diversity of marine ecosystems, as well as providing new effective ways to manage marine debris, thus ensuring a sustainable marine environment for future generations. AI-based drone image analysis systems are expected to provide modern and innovative solutions to these environmental challenges.

Appendix

- Kakao Impact presentation ('23) :
<https://www.youtube.com/watch?v=HFQcZg5kKYY&list=PLRUPfoNYGoLntVcia3ITk95zkMOKZPbCO>
 - Kakao Impact is Kakao's corporate foundation that uses technology and platforms to create valuable change.
 - This presentation is about the process and outcome of developing an AI system to help preserve the habitat of the southern bottlenose dolphin in Jeju in 2023.
 - We collaborated with Marine Animal Research and Conservation (MARC), a research-driven non-profit organization that aims to improve the ecology and conservation of marine animals.
- Related result :
<https://drive.google.com/drive/u/0/folders/12iazT0GhGXhumX7RAVwJ3niDL5N4q83Y>